Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 1 of 8 · Question 6 part b.ii · 4 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

6	(a)	(i)	any two from: • alcohol absorbed/passes into blood; • rapidly/immediately/straight away/steep increase; • by stomach;	ignore intestines	2
		(ii)	• (alcohol carried to) liver; • broken down/detoxified; • level (in blood) falls;		3
	(b)	(i)	• any time after (including) 2.4 hours/2 hours 24 mins;	allow a time of between 1-20 minutes	1
		(ii)	• alcohol is a depressant; • slows reaction/increases reaction time; • drivers can't respond as quickly/less alert; • leads to accidents;		4
	(c)		• collecting duct less permeable (in absence of ADH); • less water reabsorbed; • into blood; • volume of urine (therefore) increased/more urine produced; • and is more dilute/less concentrated;	accept ref fewer aquaporins/water channels	5
					15 marks

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 1 of 8 · Question 6 part b.ii · 4 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 2 of 8 · Question 7 part b.ii · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
7	(a)	$\frac{14}{100}$ x 250 = 35;	Ecf one mark for correct answer from wrong numbers	1
		35 ÷ 10 = 3.5	Full marks for correct final answer of 3.5	1
	(b) (i)	4 x 3.5 = 14 (units);	ecf from 7a	1
		14 – 3 = 11 (hours);	Full marks for correct final answer of 11 hours	1
	(ii)	Any two from: • alcohol is a depressant; • (alcohol) affects the nervous system/brain; • slows reactions/increases reaction time;		Max 2
	(c)	Any five from: • less/no bile produced; • less/no emulsification of fat; • surface area of fat remains low; • less fat digested/broken down (by lipase); • pH of small intestine remains low/acidic; • enzymes denatured/reduced enzyme activity; • reduced/no digestion by protease / carbohydrase/fewer proteins/carbohydrates digested/broken down; • more fat egested;	Allow trypsin for protease and amylase for carbohydrase	Max 5

Total question 7 = 11 marks

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 2 of 8 · Question 7 part b.ii · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 3 of 8 · Question 9 part a · 3 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

9 (a)	Any three of the following • causes scarring/cirrhosis; • unable to detoxify substances / produce bile; • gene mutation; • leading to cancer;		3
(b)	Any six from the following • virus attaches to (host) cell; • DNA injected into cell; • DNA becomes incorporated into host cell DNA; • viral DNA replicated along with host cell DNA; • protein synthesis; • new virus produced; • lysis of host cell; • new viruses released (to infect new cells);		6
(c)	• antigens injected; • white cells/lymphocytes produce antibodies; • memory cells produce antibodies faster;		3
			Total 12

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 3 of 8 · Question 9 part a · 3 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 4 of 8 · Question 6 part a.i · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

6	(a)	(i)	any two from: • alcohol absorbed/passes into blood; • rapidly/immediately/straight away/steep increase; • by stomach;	ignore intestines	2
		(ii)	• (alcohol carried to) liver; • broken down/detoxified; • level (in blood) falls;		3
	(b)	(i)	• any time after (including) 2.4 hours/2 hours 24 mins;	allow a time of between 1-20 minutes	1
		(ii)	• alcohol is a depressant; • slows reaction/increases reaction time; • drivers can't respond as quickly/less alert; • leads to accidents;		4
	(c)		• collecting duct less permeable (in absence of ADH); • less water reabsorbed; • into blood; • volume of urine (therefore) increased/more urine produced; • and is more dilute/less concentrated;	accept ref fewer aquaporins/water channels	5
					15 marks

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 4 of 8 · Question 6 part a.i · 2 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 5 of 8 · Question 4 part b · 3 marks

2020 NOV · Paper 02 · find part b in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 6 of 8 · Question 5 part b · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Question number			
5 (a)	• addition of figures = 1690; • divide by 10 = 169;		2
(b)	makes it faster/decreases reaction time;		1
(c)	• caffeine has to be absorbed; • into blood; • transported to nervous system/brain;	Ignore body	3
(d)	• can't control what is in food/drink; • could contain caffeine/other factors that would affect result/to ensure caffeine is the cause;	Ignore coffee	2
(e)	185 ± 5 mm;		1 Total 9

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 6 of 8 · Question 5 part b · 1 mark

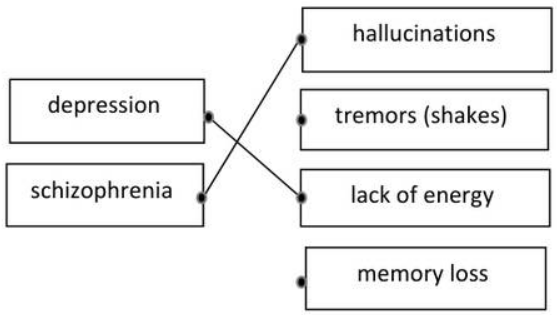
2021 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 7 of 8 · Question 2 part c · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

2 (a) (i)	2002;		1
(ii)	2005 (1.4)		1
(iii)	2.75:2.6; 1.1/1.06 :1;		2
(iv)	low income/low self esteem/(domestic) violence/low social status/substance abuse;		1
(b)			2
(c)	**A** alcohol and heroin; B antibiotics have no effect on mental health C paracetamol has no effect on mental health D antibiotics have no effect on mental health		1
(d)	Any two from the following brain/skin/sweat glands/muscles/blood vessels/arterioles;;		2
			Total 10

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 7 of 8 · Question 2 part c · 1 mark

2022 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 8 of 8 · Question 3 part b · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

3 (a)	(i)	Synapse/synaptic cleft;		1
	(ii)	arrow pointing from top to bottom;		1
	(iii)	any six from • vesicles; • contain neurotransmitter/ named neurotransmitter; • (vesicles)fuse/binds/attaches with pre-synaptic membrane/(vesicles) moves to pre-synaptic membrane; • exocytosis/release of neurotransmitter; • which diffuses (across the gap/synapse)/description of diffusion; • binds/attaches with receptors on post-synaptic membrane; • (initiates)new impulse;		6
	(iv)	• neurotransmitter only released from pre-synaptic membrane/ neurone A/first neurone/(vesicles) with neurotransmitter in pre-synaptic knob/neurone A/first neurone; • diffuses down a concentration gradient/description of concentration gradient; • receptors only on post-synaptic membrane/neurone B/ second neurone;		3
(b)		any two from • addiction; • stroke/heart problems/high blood pressure; • steal to fund habit; • anxiety; • depression; • hallucinations; • memory loss; • brain damage/ damage nerve cells; • paranoia; • overdose/death;		2

Total 13 marks

Type MS · Explain the effects and dangers of alcohol and other drugs

Answer 8 of 8 · Question 3 part b · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks